Test Case Document (Updated)

Version 2.0, 11 November 2026. Updates STP Appendix A after Sprints 1-3. Team: Krish Mathur, Atharva Mihir Gore, Samyuktha, Tanishk Gowda.

1. Purpose

This document updates the test cases in Appendix A of the Software Test Plan (STP) to match the system as built in Sprints 1 to 3, as required by Mini-Project Deliverables Part-3, item 1. It lists every test case with its expected result, the actual result and test result from the final regression run, and the automated tests that implement it. Each automated test name contains its test case ID (STP section 11).

2. Summary of changes

	Unit (UT)	Integration (IT)	System (ST)	Total
Planned in STP	28	14	16	58
Added in Sprint 3	2	4	1	7
Total now	30	18	17	65

Revised cases (15) - steps, data or expected results changed because the implementation changed during the sprints:

ID	What changed
IT_AUTH_02	Availability page path is /rooms/ (was /rooms/availability/).
ST_AUTH_01	Student landing page is Room availability; own profile is one click away in the navbar.
UT_STU_01	Form renamed StudentRegistrationForm; it also creates the student's login with a password.
UT_STU_04	Update goes through the update_student() service, which also keeps the login email in step.
UT_HOS_01	Hostel has code, total capacity, address and notes; there are no Type or Warden fields.
IT_AVL_01	Response also carries room_id, hostel and room_number (SAD 3.3.3).
UT_ALLOC_05	On SQLite the losing thread retries after a 50 ms wait when the database is locked.
IT_STAT_01	Student home now shows a 'My room' card (hostel, room, floor, dates, check-in status).
ST_ALLOC_02	Cancel has a confirmation page (SAD 3.5).
ST_CICO_01	Check-out has a confirmation page; check-in is POST only.
UT_DASH_01	Function is build_summary(); student count includes every non-staff login.
ST_DASH_01	Compared against ORM counts instead of DB Browser for SQLite.
ST_PERF_01	Run in-process with the Django test client (160 requests over 4 pages) instead of Locust.
ST_REL_01	Run as two separate Python processes on a file database (write, exit, re-open, verify).
ST_SEC_01	No logs/hms.log is configured, so the log search step does not apply.

New cases (7) - behaviour added during the sprints that the STP did not cover:

ID	Description	Requirement
UT_CI_03	Check-in on a cancelled or completed allocation is rejected	HMS-F-015
IT_CI_01	Check-in only accepts POST	HMS-F-015
UT_CO_03	Check-out twice is rejected	HMS-F-016
IT_CO_03	Check-out GET only shows the confirmation step	HMS-F-016
IT_SEC_01	Hostile strings through the real search and forms	HMS-SR-004
IT_SEC_02	Custom 403, 404 and 500 pages	HMS-SR-002 (SAD 3.4)
ST_CI_01	CI pipeline runs the suite with coverage gate	HMS-NF-005

Execution result: 64 of 65 cases passed, 0 failed, 1 not run (ST_UX_01, which needs three users who did not build HMS and is scheduled for the UAT demo).

3. Test cases

UC-01 Login and Access Control

ID	Module	Description / requirement	Pre-conditions and data	Expected result	Actual result	Result
UT_AUTH_01	AuthModule / authenticate()	Valid credentials return the user HMS-F-001	Seeded admin1 and stu1	User object returned; admin has is_staff = True	As expected; 2 automated test(s) passed	**Pass**
UT_AUTH_02	AuthModule / authenticate()	Wrong password or unknown user is rejected HMS-F-001	admin1 / wrong123	None returned; no session created	As expected; 2 automated test(s) passed	**Pass**
UT_AUTH_04	AuthModule / @admin_required	Decorator blocks non-admin users HMS-F-002, SR-002	RequestFactory requests as stu1 and admin1	Student: PermissionDenied (403); admin: 200	As expected; 2 automated test(s) passed	**Pass**
IT_AUTH_02	URLs / login_required	Protected pages need login HMS-SR-001, NF-003 *Revised: Availability page path is /rooms/ (was /rooms/availability/).*	Not logged in	302 to /accounts/login/?next=<url> for every protected page	As expected; 1 automated test(s) passed	**Pass**
IT_AUTH_03	URLs / @admin_required	Student session cannot reach admin pages HMS-F-002, SR-002	Logged in as stu1; valid allocation form data	403 for every admin GET and POST; nothing saved	As expected; 2 automated test(s) passed	**Pass**
IT_AUTH_04	AuthModule / logout	Logout ends the session HMS-SR-005	Logged in as stu1	After POST /accounts/logout/, old session is redirected to login	As expected; 1 automated test(s) passed	**Pass**
ST_AUTH_01	Login page (UI)	Login lands on the right page for each role HMS-F-001, F-002 *Revised: Student landing page is Room availability; own profile is one click away in the navbar.*	admin1, stu1; wrong123	Admin sees dashboard; student sees room availability; wrong password shows an error	As expected; 3 automated test(s) passed	**Pass**

UC-02 Student Management

ID	Module	Description / requirement	Pre-conditions and data	Expected result	Actual result	Result
UT_STU_01	StudentModule / StudentRegistrationForm	Valid student details are saved HMS-F-003 *Revised: Form renamed StudentRegistrationForm; it also creates the student's login with a password.*	SRN PES1UG24AM999, Test Student, test@pes.edu, 9876543210, year 2	Form valid; one Student row and a non-staff login named by the SRN	As expected; 1 automated test(s) passed	**Pass**
UT_STU_02	StudentModule / StudentRegistrationForm	Duplicate SRN is rejected HMS-F-003	PES1UG24AM999 exists	Form invalid with a duplicate-SRN error	As expected; 1 automated test(s) passed	**Pass**
UT_STU_03	StudentModule / forms	Invalid email and phone are rejected HMS-SR-004	Email abc, phone 98ab12	Form invalid; errors on email and phone	As expected; 1 automated test(s) passed	**Pass**
UT_STU_04	StudentModule / update_student()	Updated details are saved HMS-F-004 *Revised: Update goes through the update_student() service, which also keeps the login email in step.*	Phone 9123456780, year 3	Reloaded record has new phone and year	As expected; 1 automated test(s) passed	**Pass**
UT_STU_05	StudentModule / search_students()	Search by SRN or part of name, case-insensitive HMS-F-017	Riya Rao (AM101), Rahul Shah (AM102)	'AM101' and 'riya' find Riya Rao; 'zzzz' finds nobody	As expected; 1 automated test(s) passed	**Pass**

ID	Module	Description / requirement	Pre-conditions and data	Expected result	Actual result	Result
IT_STU_02	URLs / student home	Student sees only own profile HMS-F-005	stu1 linked to Riya Rao	200; page shows Riya Rao's details only	As expected; 1 automated test(s) passed	**Pass**
IT_STU_03	URLs / object access	Student cannot open another student's record HMS-SR-006	stu1 and stu2 exist	403 on /students/<stu2 id>/ and /edit/; no stu2 data in response	As expected; 1 automated test(s) passed	**Pass**
ST_STU_01	Students pages (UI)	Admin registers, edits and searches a student HMS-F-003, F-004, F-017	Admin logged in	Success messages; updated phone visible; search finds the student	As expected; 3 automated test(s) passed	**Pass**

UC-03 Hostel and Room Management

ID	Module	Description / requirement	Pre-conditions and data	Expected result	Actual result	Result
UT_HOS_01	HostelModule / HostelForm	Valid hostel is saved HMS-F-006 *Revised: Hostel has code, total capacity, address and notes; there are no Type or Warden fields.*	Block C, code BLC, capacity 40, notes 'Warden: Mr. Rao'	One hostel row created	As expected; 1 automated test(s) passed	**Pass**
UT_ROOM_01	RoomModule / RoomForm	Valid room is saved HMS-F-007	Block C; room C-101, floor 1, capacity 3	Room row linked to Block C	As expected; 1 automated test(s) passed	**Pass**
UT_ROOM_02	RoomModule / capacity	Capacity boundary values HMS-F-007	Capacity 0, -1, 1	0 and -1 invalid ('Capacity must be at least 1.' for 0); 1 valid	As expected; 1 automated test(s) passed	**Pass**
UT_ROOM_03	RoomModule / uniqueness	Room number unique within a hostel HMS-F-007	C-101 in Block C; Block D exists	Second C-101 in Block C rejected; C-101 in Block D allowed	As expected; 1 automated test(s) passed	**Pass**
IT_ROOM_02	URLs / room details	Room details visible to admin and student HMS-F-009	Room C-101 with 1 ACTIVE allocation	200 for both; hostel, number, capacity, status and available beds shown	As expected; 1 automated test(s) passed	**Pass**
ST_ROOM_01	Hostels and Rooms pages	Admin sets up a hostel with rooms HMS-F-006, F-007, F-009	Block E; E-1, E-2, E-3 with capacity 1, 2, 3	All three rooms listed with correct capacity and availability	As expected; 1 automated test(s) passed	**Pass**

UC-04 Room Availability

ID	Module	Description / requirement	Pre-conditions and data	Expected result	Actual result	Result
UT_AVL_02	RoomModule / available_capacity()	Active allocations reduce availability HMS-F-008	Capacity 3 with 2 ACTIVE	Returns 1	As expected; 1 automated test(s) passed	**Pass**
UT_AVL_03	RoomModule / available_capacity()	Cancelled and completed allocations are not counted HMS-F-008, NF-006	Capacity 3: 1 ACTIVE, 1 CANCELLED, 1 COMPLETED	Returns 2	As expected; 1 automated test(s) passed	**Pass**
IT_AVL_01	JSON API / availability	Availability endpoint returns correct values HMS-F-008 *Revised: Response also carries room_id, hostel and room_number (SAD 3.3.3).*	A-101 capacity 3, 2 ACTIVE; logged in as stu1	200 {"capacity": 3, "active_allocations": 2, "available": 1}	As expected; 1 automated test(s) passed	**Pass**
ST_AVL_01	Availability (UI / API)	Student view updates after admin allocates HMS-F-008	Room A-101	Available beds reduced by 1 after the allocation	As expected; 1 automated test(s) passed	**Pass**

UC-05 Room Allocation

ID	Module	Description / requirement	Pre-conditions and data	Expected result	Actual result	Result
UT_ALLOC_01	AllocationModule / allocate_room()	Allocation succeeds when a bed is free HMS-F-010	Unallocated student; room capacity 2	ACTIVE allocation; one CREATED history row; availability 1	As expected; 2 automated test(s) passed	**Pass**
UT_ALLOC_02	AllocationModule / allocate_room()	Student with an active allocation is rejected HMS-F-010	Student already ACTIVE	AlreadyAllocatedError; no new rows	As expected; 1 automated test(s) passed	**Pass**
UT_ALLOC_03	AllocationModule / allocate_room()	Full room is rejected HMS-F-011	Capacity 2 with 2 ACTIVE	RoomFullError; active count stays 2	As expected; 2 automated test(s) passed	**Pass**
UT_ALLOC_04	AllocationModule / allocate_room()	Boundary: last bed HMS-F-011	Capacity 1; students A and B	A succeeds; B raises RoomFullError	As expected; 1 automated test(s) passed	**Pass**
UT_ALLOC_05	AllocationModule / concurrency	Two requests for the last bed at the same time HMS-F-011, NF-006 *Revised: On SQLite the losing thread retries after a 50 ms wait when the database is locked.*	1 free bed; two threads	Exactly one succeeds, the other gets RoomFullError	As expected; 1 automated test(s) passed	**Pass**
IT_ALLOC_01	URLs + AllocationModule	Allocate through the view HMS-F-010	admin1; free room; unallocated student	302 with 'Room allocated successfully'	As expected; 2 automated test(s) passed	**Pass**
IT_ALLOC_02	URLs + AllocationModule	Allocate to a full room through the view HMS-F-011	Full room	200 with 'Room has reached its capacity'; nothing saved	As expected; 1 automated test(s) passed	**Pass**
ST_ALLOC_01	Allocate page	Admin fills a room until it is full HMS-F-010, F-011	Room capacity 2	First two succeed; third shows capacity error; room shows 'Full'	As expected; 1 automated test(s) passed	**Pass**

UC-06 View, Modify and Cancel Allocation, History

ID	Module	Description / requirement	Pre-conditions and data	Expected result	Actual result	Result
UT_MOD_01	AllocationModule / modify_allocation()	Move student to a room with a free bed HMS-F-013	ACTIVE in room A; room B free	Room = B; A +1, B -1; MODIFIED history with old and new room	As expected; 1 automated test(s) passed	**Pass**
UT_MOD_02	AllocationModule / modify_allocation()	Move to a full room is rejected HMS-F-013	Room B full	RoomFullError; student stays in room A	As expected; 1 automated test(s) passed	**Pass**
UT_CAN_01	AllocationModule / cancel_allocation()	Cancel frees the bed HMS-F-014	ACTIVE allocation	Status CANCELLED; CANCELLED history; availability +1	As expected; 1 automated test(s) passed	**Pass**
UT_HIS_01	AllocationModule / history	History records every action HMS-F-018	Created, modified, cancelled by admin1	3 rows CREATED, MODIFIED, CANCELLED with performed_by admin1	As expected; 1 automated test(s) passed	**Pass**
IT_STAT_01	URLs / student home	Student sees allocation status HMS-F-012 *Revised: Student home now shows a 'My room' card (hostel, room, floor, dates, check-in status).*	stu1 allocated; stu2 not	stu1: hostel, room, status; stu2: 'No room allocated yet'	As expected; 2 automated test(s) passed	**Pass**
IT_HIS_01	URLs / history	History page filter and access HMS-F-018	History rows exist	Admin: filtered rows; student: 403	As expected; 2 automated test(s) passed	**Pass**
ST_ALLOC_02	Allocation pages	Admin modifies, cancels and checks history HMS-F-013, F-014, F-018 *Revised: Cancel has a confirmation page (SAD 3.5).*	ACTIVE allocation	Message at each step; history shows all three actions	As expected; 1 automated test(s) passed	**Pass**

UC-07 Check-in and Check-out

ID	Module	Description / requirement	Pre-conditions and data	Expected result	Actual result	Result
UT_CI_01	AllocationModule / check_in()	Check in an allocated student HMS-F-015	ACTIVE, not checked in	checked_in_on set; CHECKED_IN history row	As expected; 1 automated test(s) passed	**Pass**
UT_CI_02	AllocationModule / check_in()	Check-in twice is rejected HMS-F-015	Already checked in	InvalidStateError; date unchanged	As expected; 1 automated test(s) passed	**Pass**
UT_CI_03	AllocationModule / check_in()	Check-in on a cancelled or completed allocation is rejected HMS-F-015 **New in Sprint 3**	CANCELLED / COMPLETED allocation	InvalidStateError 'Allocation is not active'; no history	As expected; 1 automated test(s) passed	**Pass**
IT_CI_01	URLs / check-in	Check-in only accepts POST HMS-F-015 **New in Sprint 3**	Logged in as admin1	GET /allocations/<id>/checkin/ returns 405; nothing changes	As expected; 1 automated test(s) passed	**Pass**
UT_CO_01	AllocationModule / check_out()	Check-out frees the bed HMS-F-016	ACTIVE, checked in	COMPLETED; checked_out_on set; CHECKED_OUT history; availability +1	As expected; 1 automated test(s) passed	**Pass**
UT_CO_02	AllocationModule / check_out()	Check-out before check-in is rejected HMS-F-016	ACTIVE, not checked in	InvalidStateError 'Student is not checked in'	As expected; 1 automated test(s) passed	**Pass**
UT_CO_03	AllocationModule / check_out()	Check-out twice is rejected HMS-F-016 **New in Sprint 3**	Already checked out	InvalidStateError 'Allocation is not active'; one CHECKED_OUT row	As expected; 1 automated test(s) passed	**Pass**
IT_CO_01	URLs + Check-Out	Check-out updates availability HMS-F-016	Checked-in allocation in a full room	302 to allocation detail with 'Student checked out'; available = 1	As expected; 1 automated test(s) passed	**Pass**
IT_CO_02	URLs / @admin_required	Student cannot check out HMS-F-002, F-016	Logged in as stu1	403; allocation unchanged	As expected; 1 automated test(s) passed	**Pass**
IT_CO_03	URLs / check-out	Check-out GET only shows the confirmation step HMS-F-016 **New in Sprint 3**	Checked-in allocation	Confirmation page shown; allocation still ACTIVE	As expected; 1 automated test(s) passed	**Pass**
ST_CICO_01	Allocation detail (UI)	Full stay: allocate, check in, check out HMS-F-015, F-016 *Revised: Check-out has a confirmation page; check-in is POST only.*	Admin logged in	Buttons enabled only when allowed; final status COMPLETED; bed free	As expected; 1 automated test(s) passed	**Pass**

UC-08 Administrative Summary

ID	Module	Description / requirement	Pre-conditions and data	Expected result	Actual result	Result
UT_DASH_01	DashboardModule / build_summary()	Summary counts match data HMS-F-019 *Revised: Function is build_summary(); student count includes every non-staff login.*	25 students, 12 rooms, 24 beds, 15 ACTIVE	students 25, rooms 12, beds 24, occupied 15, free 9	As expected; 1 automated test(s) passed	**Pass**
IT_DASH_01	URLs / dashboard	Dashboard access HMS-F-019, F-002	Seed data	Admin: 200 with UT_DASH_01 values; student: 403	As expected; 1 automated test(s) passed	**Pass**

ID	Module	Description / requirement	Pre-conditions and data	Expected result	Actual result	Result
ST_DASH_01	Dashboard (UI)	Dashboard matches the database HMS-F-019 *Revised: Compared against ORM counts instead of DB Browser for SQLite.*	Seed data	Every value on the page equals a direct database count	As expected; 1 automated test(s) passed	**Pass**

Non-functional and Security

ID	Module	Description / requirement	Pre-conditions and data	Expected result	Actual result	Result
ST_PERF_01	Whole system	90% of pages within 3 s HMS-NF-001 *Revised: Run in-process with the Django test client (160 requests over 4 pages) instead of Locust.*	500 students, 100 rooms, 250 ACTIVE; DEBUG=False	90th percentile <= 3 s; 0 server errors	As expected; 1 automated test(s) passed	**Pass**
ST_REL_01	Whole system / database	Data survives restart HMS-NF-002 *Revised: Run as two separate Python processes on a file database (write, exit, re-open, verify).*	New student, room, allocation and check-in	All records present and unchanged after restart	Process 1 wrote student PES1UG24AM777, room REL-R-1, ACTIVE checked-in allocation and 2 history rows, then exited. Process 2 re-opened the database: all records, check-in and password hash unchanged.	**Pass**
ST_SEC_01	AuthModule / storage	Passwords stored hashed HMS-SR-003 *Revised: No logs/hms.log is configured, so the log search step does not apply.*	Users exist	Values start with pbkdf2_sha256$; no plaintext password	As expected; 2 automated test(s) passed	**Pass**
ST_SEC_02	All admin URLs	Broken access control check HMS-SR-002, SR-006	admin1, stu1, stu2	Every admin URL and foreign record returns 403 or 404	As expected; 1 automated test(s) passed	**Pass**
ST_SEC_04	Forms and search	Injection, bad input and CSRF HMS-SR-004	' OR 1=1 --, <script>, 500 chars; POST without token	Rejected or stored as text; no 500; no CSRF token gives 403	As expected; 3 automated test(s) passed	**Pass**
IT_SEC_01	Search + forms (end to end)	Hostile strings through the real search and forms HMS-SR-004 **New in Sprint 3**	' OR 1=1 --, <script>, 500 chars as admin1	200/302 only; search returns no extra records; no unescaped script in any page	As expected; 2 automated test(s) passed	**Pass**
IT_SEC_02	Error pages	Custom 403, 404 and 500 pages HMS-SR-002 (SAD 3.4) **New in Sprint 3**	stu1 on admin page; unknown URL and record; forced 500	Custom page for each code; no stack trace	As expected; 4 automated test(s) passed	**Pass**
ST_UX_01	UI	Usability walk-through HMS-NF-004	3 students who did not build HMS	All tasks finished without help; messages rated clear	Needs three students who did not build HMS. To be run during the UAT demo.	**Not run**
ST_MAINT_01	Code structure	Modular structure HMS-NF-005	Source on main	One app per module; full suite passes after a change	As expected; 1 automated test(s) passed	**Pass**
ST_DATA_01	Room + Allocation	Capacity and availability stay consistent HMS-NF-006	Random allocate/modify/cancel/check-out actions	ACTIVE <= capacity; availability = capacity - ACTIVE	As expected; 1 automated test(s) passed	**Pass**

ID	Module	Description / requirement	Pre-conditions and data	Expected result	Actual result	Result
ST_CI_01	GitHub Actions	CI pipeline runs the suite with coverage gate HMS-NF-005 **New in Sprint 3**	Push / PR to main	Workflow passes on Python 3.11 and 3.13 with coverage >= 80%	Workflow commands (makemigrations --check, coverage run --branch, coverage report --fail-under=80) pass locally: 202 tests, 99%. Confirm the green run on the GitHub Actions tab.	**Pass (local)**

4. Updated traceability (requirement to test cases)

Requirement	Name	Test cases	Status
HMS-F-001	User authentication	UT_AUTH_01, UT_AUTH_02, ST_AUTH_01	Pass
HMS-F-002	Role-based access control	UT_AUTH_04, IT_AUTH_03, ST_AUTH_01, IT_CO_02, IT_DASH_01	Pass
HMS-F-003	Student registration	UT_STU_01, UT_STU_02, ST_STU_01	Pass
HMS-F-004	Update student details	UT_STU_04, ST_STU_01	Pass
HMS-F-005	View student profile	IT_STU_02	Pass
HMS-F-006	Hostel management	UT_HOS_01, ST_ROOM_01	Pass
HMS-F-007	Room inventory and management	UT_ROOM_01, UT_ROOM_02, UT_ROOM_03, ST_ROOM_01	Pass
HMS-F-008	Room availability tracking	UT_AVL_02, UT_AVL_03, IT_AVL_01, ST_AVL_01	Pass
HMS-F-009	Room search and view	IT_ROOM_02, ST_ROOM_01	Pass
HMS-F-010	Room allocation	UT_ALLOC_01, UT_ALLOC_02, IT_ALLOC_01, ST_ALLOC_01	Pass
HMS-F-011	Prevent over-allocation	UT_ALLOC_03, UT_ALLOC_04, UT_ALLOC_05, IT_ALLOC_02, ST_ALLOC_01	Pass
HMS-F-012	View allocation status	IT_STAT_01	Pass
HMS-F-013	Modify room allocation	UT_MOD_01, UT_MOD_02, ST_ALLOC_02	Pass
HMS-F-014	Cancel room allocation	UT_CAN_01, ST_ALLOC_02	Pass
HMS-F-015	Student check-in	UT_CI_01, UT_CI_02, UT_CI_03, IT_CI_01, ST_CICO_01	Pass
HMS-F-016	Student check-out	UT_CO_01, UT_CO_02, UT_CO_03, IT_CO_01, IT_CO_02, IT_CO_03, ST_CICO_01	Pass
HMS-F-017	Search student records	UT_STU_05, ST_STU_01	Pass
HMS-F-018	Allocation history log	UT_HIS_01, IT_HIS_01, ST_ALLOC_02	Pass
HMS-F-019	Administrative summary dashboard	UT_DASH_01, IT_DASH_01, ST_DASH_01	Pass
HMS-NF-001	Performance (90% of pages within 3 s)	ST_PERF_01	Pass
HMS-NF-002	Reliability and persistence	ST_REL_01	Pass
HMS-NF-003	Security (mandatory authentication)	IT_AUTH_02	Pass
HMS-NF-004	Usability and clear messages	ST_UX_01	Partly run
HMS-NF-005	Maintainability (modular apps)	ST_MAINT_01, ST_CI_01	Pass
HMS-NF-006	Data integrity of capacity	UT_AVL_03, UT_ALLOC_05, ST_DATA_01	Pass
HMS-SR-001	Mandatory authentication	IT_AUTH_02	Pass
HMS-SR-002	Role-based authorisation	UT_AUTH_04, IT_AUTH_03, ST_SEC_02, IT_SEC_02	Pass
HMS-SR-003	Password hashing	ST_SEC_01	Pass
HMS-SR-004	Input validation and sanitisation	UT_STU_03, ST_SEC_04, IT_SEC_01	Pass
HMS-SR-005	Session management and logout	IT_AUTH_04	Pass
HMS-SR-006	Student data privacy	IT_STU_03, ST_SEC_02	Pass

5. Automated test reference

Test case ID to the automated test(s) that implement it, from the final run.

ID	Automated test(s)
UT_AUTH_01	accounts.tests.AuthenticationModuleTests.test_TC_AUTH_01_UT_AUTH_01_valid_credentials_return_admin_user accounts.tests.AuthenticationModuleTests.test_TC_AUTH_01_UT_AUTH_01_valid_credentials_return_student_user
UT_AUTH_02	accounts.tests.AuthenticationModuleTests.test_TC_AUTH_01_UT_AUTH_02_nonexistent_user_rejected accounts.tests.AuthenticationModuleTests.test_TC_AUTH_01_UT_AUTH_02_wrong_password_rejected
UT_AUTH_04	accounts.tests.RoleBasedAccessControlTests.test_TC_AUTH_02_UT_AUTH_04_admin_required_allows_admin accounts.tests.RoleBasedAccessControlTests.test_TC_AUTH_02_UT_AUTH_04_admin_required_blocks_student
IT_AUTH_02	accounts.tests.MandatoryAuthSecurityTests.test_TC_SEC_01_IT_AUTH_02_protected_routes_require_authentication
IT_AUTH_03	accounts.tests.RoleAuthorizationAdminProtectionTests.test_TC_SEC_02_IT_AUTH_03_ST_SEC_02_student_cannot_access_any_admin_page accounts.tests.RoleAuthorizationAdminProtectionTests.test_TC_SEC_02_IT_AUTH_03_student_cannot_post_to_admin_creation_endpoints
IT_AUTH_04	accounts.tests.SessionSecurityManagementTests.test_TC_SEC_05_IT_AUTH_04_logout_invalidates_session_and_clears_cookie
ST_AUTH_01	accounts.tests.AuthenticationModuleTests.test_TC_AUTH_01_ST_AUTH_01_admin_login_lands_on_dashboard accounts.tests.AuthenticationModuleTests.test_TC_AUTH_01_ST_AUTH_01_student_login_lands_on_rooms accounts.tests.AuthenticationModuleTests.test_TC_AUTH_01_ST_AUTH_01_wrong_password_displays_error
UT_STU_01	students.tests.StudentFormTests.test_TC_STU_01_UT_STU_01_valid_details_are_saved
UT_STU_02	students.tests.StudentFormTests.test_TC_STU_01_UT_STU_02_duplicate_srn_is_rejected
UT_STU_03	students.tests.StudentFormTests.test_TC_STU_01_UT_STU_03_invalid_email_and_phone_are_rejected
UT_STU_04	students.tests.StudentFormTests.test_TC_STU_02_UT_STU_04_updated_details_are_saved
UT_STU_05	students.tests.StudentSearchTests.test_TC_STU_04_UT_STU_05_search_by_srn_or_name
IT_STU_02	students.tests.StudentViewTests.test_TC_STU_03_IT_STU_02_student_sees_only_own_profile
IT_STU_03	students.tests.StudentViewTests.test_TC_SEC_06_IT_STU_03_student_cannot_open_another_students_record
ST_STU_01	students.tests.StudentViewTests.test_TC_STU_01_ST_STU_01_admin_registers_student students.tests.StudentViewTests.test_TC_STU_02_ST_STU_01_admin_edits_student students.tests.StudentViewTests.test_TC_STU_04_ST_STU_01_admin_searches_students
UT_HOS_01	hostels.tests.HostelFormUnitTests.test_TC_HOSTEL_01_UT_HOS_01_valid_hostel_is_saved
UT_ROOM_01	rooms.tests.RoomDesignedCaseTests.test_TC_ROOM_01_UT_ROOM_01_valid_room_is_saved
UT_ROOM_02	rooms.tests.RoomDesignedCaseTests.test_TC_ROOM_01_UT_ROOM_02_capacity_boundary_values
UT_ROOM_03	rooms.tests.RoomDesignedCaseTests.test_TC_ROOM_01_UT_ROOM_03_room_number_unique_within_hostel
IT_ROOM_02	rooms.tests.RoomDesignedCaseTests.test_TC_ROOM_03_IT_ROOM_02_room_details_visible_to_admin_and_student
ST_ROOM_01	rooms.tests.RoomDesignedCaseTests.test_TC_ROOM_01_TC_ROOM_03_ST_ROOM_01_admin_sets_up_hostel_with_rooms
UT_AVL_02	rooms.tests.RoomDesignedCaseTests.test_TC_ROOM_02_UT_AVL_02_active_allocations_reduce_availability
UT_AVL_03	rooms.tests.RoomDesignedCaseTests.test_TC_ROOM_02_UT_AVL_03_cancelled_and_completed_are_not_counted
IT_AVL_01	rooms.test_api.RoomAvailabilityApiTests.test_TC_ROOM_02_IT_AVL_01_returns_correct_values_for_student
ST_AVL_01	rooms.test_api.RoomAvailabilityApiTests.test_TC_ROOM_02_ST_AVL_01_value_updates_after_allocation

ID	Automated test(s)
UT_ALLOC_01	allocations.tests.AllocateRoomServiceTests.test_TC_ALLOC_01_UT_ALLOC_01_allocate_to_empty_room allocations.tests.AllocationModelUnitTests.test_TC_ALLOC_01_UT_ALLOC_01_allocation_creation_with_defaults
UT_ALLOC_02	allocations.tests.AllocateRoomServiceTests.test_TC_ALLOC_01_UT_ALLOC_02_student_already_allocated_rejected
UT_ALLOC_03	allocations.tests.AllocationCapacityIntegrationTests.test_TC_ALLOC_02_UT_ALLOC_03_cancelled_allocations_do_not_reduce_capacity allocations.tests.AllocateRoomServiceTests.test_TC_ALLOC_02_UT_ALLOC_03_full_room_rejected
UT_ALLOC_04	allocations.tests.AllocateRoomServiceTests.test_TC_ALLOC_02_UT_ALLOC_04_capacity_one_second_student_rejected
UT_ALLOC_05	allocations.tests.AllocateRoomConcurrencyTests.test_TC_ALLOC_02_TC_DATA_01_UT_ALLOC_05_two_threads_one_bed
IT_ALLOC_01	allocations.tests.AllocationCapacityIntegrationTests.test_TC_ALLOC_01_IT_ALLOC_01_active_allocation_reduces_room_capacity allocations.tests.AllocationViewTests.test_TC_ALLOC_01_IT_ALLOC_01_admin_allocates_through_view
IT_ALLOC_02	allocations.tests.AllocationViewTests.test_TC_ALLOC_02_IT_ALLOC_02_full_room_shows_error_and_saves_nothing
ST_ALLOC_01	allocations.tests.AllocationViewTests.test_TC_ALLOC_02_ST_ALLOC_01_fill_room_through_views_then_third_fails
UT_MOD_01	allocations.tests.ModifyAndCancelServiceTests.test_TC_ALLOC_04_UT_MOD_01_modify_to_room_with_free_bed
UT_MOD_02	allocations.tests.ModifyAndCancelServiceTests.test_TC_ALLOC_04_UT_MOD_02_modify_to_full_room_rejected
UT_CAN_01	allocations.tests.ModifyAndCancelServiceTests.test_TC_ALLOC_05_UT_CAN_01_cancel_frees_bed_and_keeps_row
UT_HIS_01	allocations.tests.ModifyAndCancelServiceTests.test_TC_ALLOC_06_UT_HIS_01_history_records_each_change_by_admin
IT_STAT_01	students.tests.StudentHomeCardTests.test_TC_ALLOC_03_IT_STAT_01_card_shows_room_details_and_awaiting_check_in allocations.tests.AllocationViewTests.test_TC_ALLOC_03_IT_STAT_01_student_home_shows_allocation_or_none
IT_HIS_01	allocations.tests.AllocationViewTests.test_TC_ALLOC_06_IT_HIS_01_history_filtered_by_student allocations.tests.AllocationViewTests.test_TC_ALLOC_06_IT_HIS_01_student_gets_403_on_history
ST_ALLOC_02	allocations.tests.AllocationViewTests.test_TC_ALLOC_04_TC_ALLOC_05_TC_ALLOC_06_ST_ALLOC_02_modify_cancel_history_flow
UT_CI_01	allocations.test_checkin_checkout.CheckInServiceTests.test_TC_CI_01_UT_CI_01_check_in_sets_date_and_history
UT_CI_02	allocations.test_checkin_checkout.CheckInServiceTests.test_TC_CI_01_UT_CI_02_check_in_twice_is_rejected
UT_CI_03	allocations.test_checkin_checkout.CheckInServiceTests.test_TC_CI_01_UT_CI_03_check_in_on_inactive_allocation_is_rejected
IT_CI_01	allocations.test_checkin_checkout.CheckInOutViewTests.test_TC_CI_01_IT_CI_01_check_in_requires_post
UT_CO_01	allocations.test_checkin_checkout.CheckOutServiceTests.test_TC_CO_01_UT_CO_01_check_out_completes_and_frees_bed
UT_CO_02	allocations.test_checkin_checkout.CheckOutServiceTests.test_TC_CO_01_UT_CO_02_check_out_before_check_in_is_rejected
UT_CO_03	allocations.test_checkin_checkout.CheckOutServiceTests.test_TC_CO_01_UT_CO_03_check_out_twice_is_rejected
IT_CO_01	allocations.test_checkin_checkout.CheckInOutViewTests.test_TC_CO_01_IT_CO_01_check_out_through_url_updates_availability
IT_CO_02	allocations.test_checkin_checkout.CheckInOutViewTests.test_TC_CO_01_IT_CO_02_student_cannot_check_in_or_out
IT_CO_03	allocations.test_checkin_checkout.CheckInOutViewTests.test_TC_CO_01_IT_CO_03_check_out_get_shows_confirmation_only
ST_CICO_01	allocations.test_checkin_checkout.CheckInOutSystemTests.test_TC_CI_01_TC_CO_01_ST_CICO_01_full_stay
UT_DASH_01	dashboard.tests.DashboardDesignedCaseTests.test_TC_DASH_01_UT_DASH_01_summary_counts_match_data
IT_DASH_01	dashboard.tests.DashboardDesignedCaseTests.test_TC_DASH_01_IT_DASH_01_dashboard_access_by_role
ST_DASH_01	dashboard.tests.DashboardDesignedCaseTests.test_TC_DASH_01_ST_DASH_01_displayed_values_match_database
ST_PERF_01	dashboard.tests.LoadTests.test_TC_PERF_01_ST_PERF_01_ninetieth_percentile_under_three_seconds

ID	Automated test(s)
ST_REL_01	Process 1 wrote student PES1UG24AM777, room REL-R-1, ACTIVE checked-in allocation and 2 history rows, then exited. Process 2 re-opened the database: all records, check-in and password hash unchanged.
ST_SEC_01	accounts.tests.PasswordSecurityStorageTests.test_TC_SEC_03_ST_SEC_01_different_users_with_same_password_have_unique_hashes accounts.tests.PasswordSecurityStorageTests.test_TC_SEC_03_ST_SEC_01_passwords_stored_hashed_with_pbkdf2
ST_SEC_02	accounts.tests.RoleAuthorizationAdminProtectionTests.test_TC_SEC_02_IT_AUTH_03_ST_SEC_02_student_cannot_access_any_admin_page
ST_SEC_04	accounts.tests.InputSanitizationAndValidationTests.test_TC_SEC_04_ST_SEC_04_500_character_fuzzing_boundary accounts.tests.InputSanitizationAndValidationTests.test_TC_SEC_04_ST_SEC_04_csrf_token_required_on_post accounts.tests.InputSanitizationAndValidationTests.test_TC_SEC_04_ST_SEC_04_xss_payloads_rejected
IT_SEC_01	accounts.tests.EndToEndFuzzingTests.test_TC_SEC_04_IT_SEC_01_hostile_form_input_is_rejected_or_stored_as_text accounts.tests.EndToEndFuzzingTests.test_TC_SEC_04_IT_SEC_01_search_returns_no_extra_records_and_no_error
IT_SEC_02	accounts.tests.ErrorPageTests.test_TC_SEC_02_IT_SEC_02_server_error_page_has_no_stack_trace accounts.tests.ErrorPageTests.test_TC_SEC_02_IT_SEC_02_student_on_admin_page_gets_custom_403 accounts.tests.ErrorPageTests.test_TC_SEC_02_IT_SEC_02_unknown_record_gets_custom_404 accounts.tests.ErrorPageTests.test_TC_SEC_02_IT_SEC_02_unknown_url_gets_custom_404
ST_UX_01	Needs three students who did not build HMS. To be run during the UAT demo.
ST_MAINT_01	accounts.tests.ModularArchitectureMaintenanceTests.test_TC_MAINT_01_ST_MAINT_01_installed_apps_decoupled
ST_DATA_01	allocations.tests.CapacityConsistencySystemTests.test_TC_DATA_01_ST_DATA_01_random_actions_keep_capacity_consistent
ST_CI_01	Workflow commands (makemigrations --check, coverage run --branch, coverage report --fail-under=80) pass locally: 202 tests, 99%. Confirm the green run on the GitHub Actions tab.